

Breakeven Analysis

Topic 7 — MBA Financial Management Accounting

A comprehensive guide to Cost–Volume–Profit analysis: understanding how sales volume, pricing, variable costs, and fixed costs interact to determine profitability and inform strategic business decisions.



What Is Breakeven Analysis?

Core Definition

Break-even analysis (Cost–Profit Analysis) is a management accounting technique used to determine the level of sales or output at which **total revenue equals total costs**, resulting in neither a profit nor a loss.

It examines the relationship between sales volume, selling price, variable costs, fixed costs, and profit to determine the break-even point and assess how changes in these factors affect profitability.

Key Applications

- Pricing decisions — setting optimal selling prices
- Profit planning — forecasting required sales volumes
- Cost control — understanding cost structure impact
- Determining the sales volume required to achieve a desired profit level

Also Known As

C-V-P Analysis (Cost–Volume–Profit Analysis) a broader term that captures the full relationship between these financial drivers.

Key Questions C-V-P Analysis Answers

1

Break-Even Volume

How many units must be sold to break even i.e., make no profit or loss?
For a service organization (e.g., transport), how many passengers must be carried to avoid a loss?

2

Target Revenue

How many units or how much revenue must the company generate to achieve a specific target profit level (Targeted Profit)?

3

Margin of Safety

How safe are we before we make a loss? $MOS = \text{Target Revenue} - \text{BEP Revenue}$. This measures the buffer between current/target sales and the break-even point.

4

Sensitivity

How sensitive is the breakeven point to changes in costs and selling prices? Understanding this helps management assess risk and make informed decisions.



The Five Breakeven Assumptions

Breakeven analysis rests on a set of simplifying assumptions. Understanding these is critical to applying the technique correctly and recognizing its limitations.

1 Cost Separability

All costs in the business can be split into variable and fixed costs, there are no mixed costs in the model.

2 Constant Prices and Variable Costs

Selling price per unit and variable cost per unit remain constant throughout the relevant range of output.

3 Constant Fixed Costs

Total fixed costs remain constant and there is no change in the relevant range of production or sales.

4 No Inventory Changes

All goods produced or purchased will be sold, there are no opening or closing inventory balances to consider.

5 No Productivity Changes

Technology and productivity remain the same throughout the period, no efficiency gains or losses are assumed.

Deriving the Breakeven Formula

The breakeven formula is built from the fundamental profit equation. Starting from first principles:

1

Step 1: Profit Equation

Profit = Total Revenue – Total Costs

Total Costs = Total Variable Costs + Total Fixed Costs

2

Step 2: Expand

$\Pi = (SP \times Q) - (VC \times Q) - TFC$

Revenue = $SP \times Q$; Variable Costs = $VC \times Q$

3

Step 3: Factor

$\Pi = (SP - VC) \times Q - TFC$

($SP - VC$) is defined as **Contribution per Unit**

4

Step 4: Set $\Pi = 0$

$QBE = TFC \div (SP - VC)$

At breakeven, profit = 0, so $TFC = \text{Contribution} \times Q$

📌 The term **Contribution** ($SP - VC$) is central to all breakeven calculations. It represents the amount each unit sold contributes toward covering fixed costs and generating profit.

The Root Breakeven Formula — Symbol Reference

The Master Formula

$$\Pi = (SP \times Q) - (VC \times Q) - TFC$$

This is the root breakeven formula. Any component can be made the subject of the formula using algebra to solve for that unknown.

Break-Even Units: $Q_{BE} = TFC \div (SP - VC)$

Break-Even Revenue: $BE \text{ Revenue} = Q_{BE} \times SP$

Symbol Definitions

- Π — Profit Before Tax
- Q — Quantities (units sold)
- SP — Selling Price per unit
- VC — Variable Cost per unit
- TFC — Total Fixed Costs
- $(SP - VC)$ — Contribution per unit

After-Tax Adjustment

If a target profit after tax (PAT) is given, gross it up before using the formula: **$PBT = PAT \div (1 - \text{Tax Rate})$**

Rearranging the Formula for Each Component

By making each variable the subject of the root formula, management can solve for any unknown given the others. The table below summarizes all key rearrangements:

Component	Meaning	Formula
Profit Before Tax (Π)	Target profit before tax; if PAT is given, gross up by dividing by $(1 - \text{Tax Rate})$	$\Pi = (SP \times Q) - (VC \times Q) - TFC$
Selling Price (SP)	The selling price required per unit to achieve a given profit	$SP = [\Pi + (VC \times Q) + TFC] \div Q$
Quantity (Q)	Units that must be sold to achieve a given profit (Target Sales in units)	$Q = (\Pi + TFC) \div (SP - VC)$
Variable Cost (VC)	Maximum variable cost per unit consistent with a given profit	$VC = [(SP \times Q) - TFC - \Pi] \div Q$
Total Fixed Costs (TFC)	Maximum fixed costs that can be incurred while achieving a given profit	$TFC = (SP \times Q) - (VC \times Q) - \Pi$

The Breakeven Graph

How to Construct the Graph

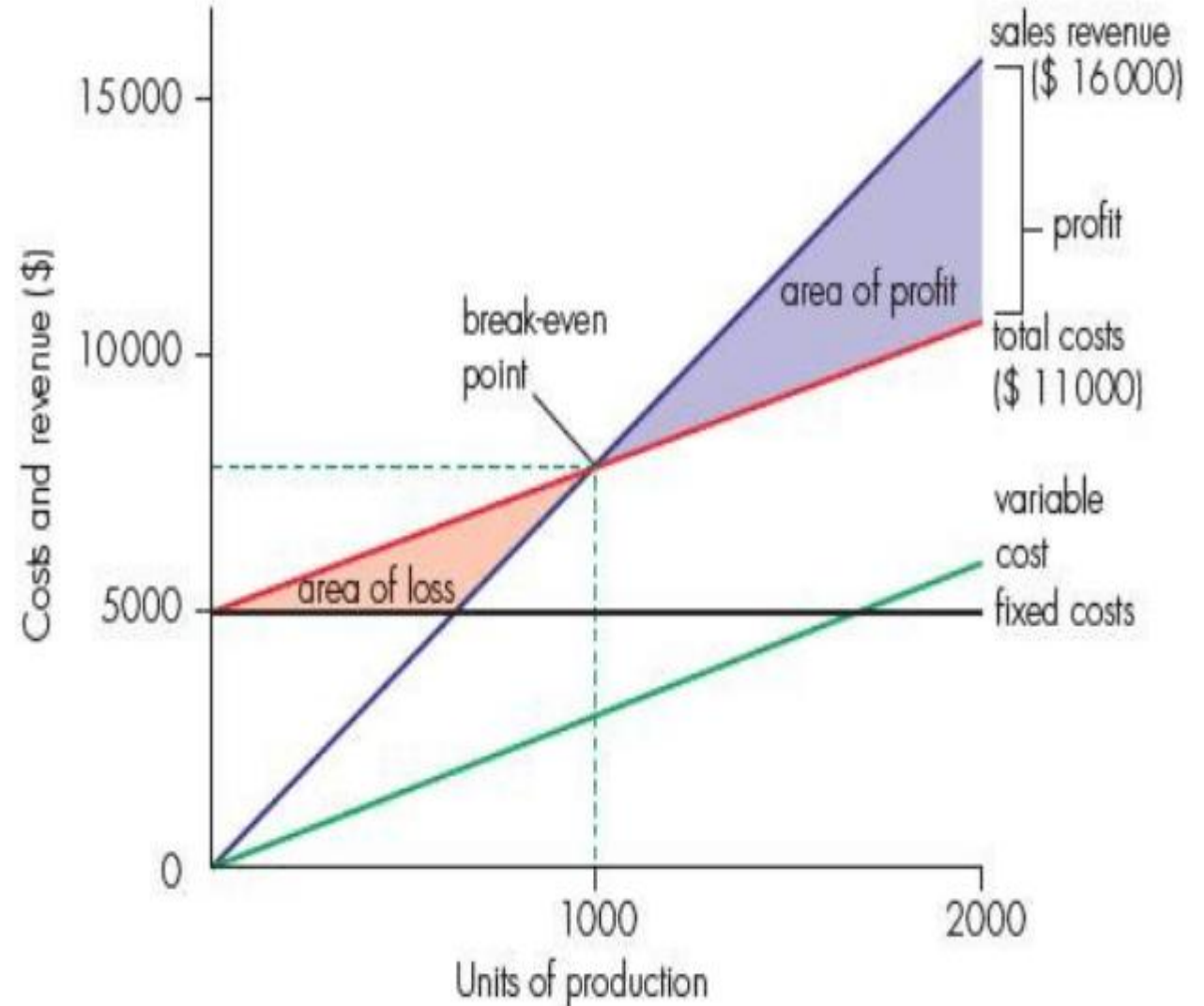
Plot **output/sales volume** on the horizontal (X) axis and **revenue and costs** on the vertical (Y) axis. Then draw three lines:

- **Total Fixed Cost line** — horizontal, constant regardless of output
- **Total Cost line** — starts at the fixed cost level and rises with variable costs as output increases
- **Total Revenue line** — starts from zero and rises according to the selling price per unit

The Break-Even Point

The point where the **Total Revenue line intersects the Total Cost line** is the break-even point. At this level of output, total revenue equals total cost the business makes zero profit and zero loss.

Output to the left of the BEP represents a **loss zone**; output to the right represents a **profit zone**.



The Operating Statement with Contribution Format

In breakeven analysis, the statement of profit or loss is restructured to highlight **contribution** rather than gross profit. This format makes the relationship between costs, volume, and profit explicit.

Sales Revenue

$SP \times \text{Quantity Sold}$

Less: Variable Costs

$VC \times \text{Quantity Sold}$

= Contribution

$\text{Revenue} - \text{Variable Costs}$

Less: Fixed Costs

Total Fixed Costs (constant)

= Profit Before Tax

$\text{Contribution} - \text{TFC}$

Less: Tax

$\text{PBT} \times \text{Tax Rate}$

📌 **Key principle:** The LOWER the breakeven point, the better for business. A higher BEP means the business must sell more to avoid losses. Higher selling prices reduce BEP; higher variable costs and fixed costs increase BEP.

Multi-Product Breakeven: Allocation of Fixed Costs

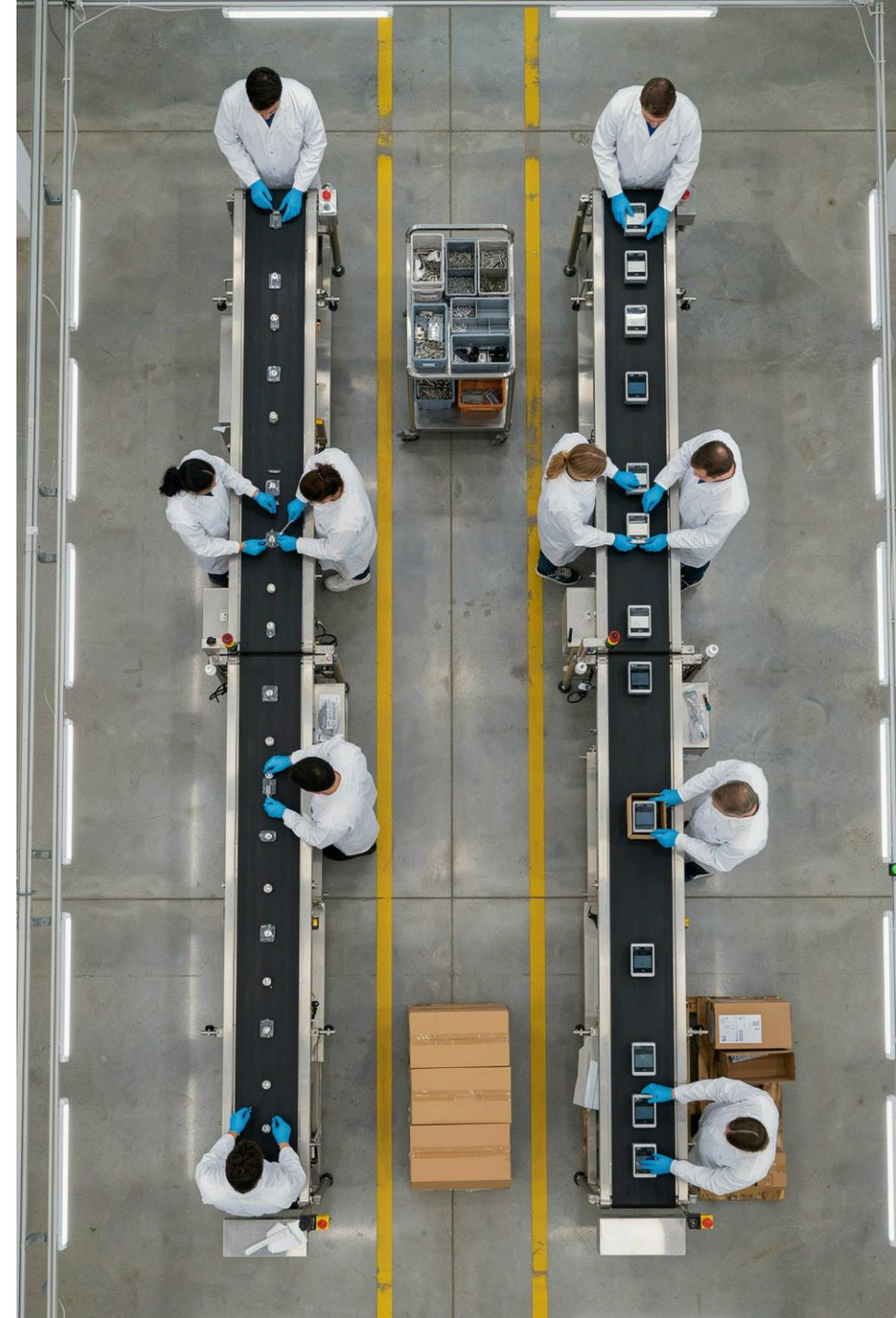
When a business sells more than one product, fixed costs must be handled carefully before computing individual breakeven points.

Approach 1: Allocate Fixed Costs

If fixed costs can be allocated to each product on a **reasonable basis** (e.g., floor space, machine hours, revenue proportion), allocate them and compute the breakeven point for each product independently using: $QBE = \text{Allocated TFC} \div \text{Contribution per Unit}$.

Approach 2: Weighted Average Contribution

If fixed costs **cannot be meaningfully allocated**, use the product mix (ratio of production/sales) to calculate a **weighted average contribution per composite unit**. Divide total TFC by this weighted average to get composite break-even units, then apportion to each product by mix ratio.



Weighted Average Contribution — Deep Dive

When a business cannot allocate fixed costs to individual products, the weighted average contribution method provides a rigorous alternative. The method works as follows:

01

Determine the Product Mix

Establish the ratio of production or sales for each product.
In the example: Product A = 40%, Product B = 60%.

02

Calculate Weighted Average Contribution

$WAC = \Sigma(\text{Mix \%} \times \text{Contribution per Unit})$. Example: $(40\% \times \text{Sh } 40) + (60\% \times \text{Sh } 60) = \text{Sh } 16 + \text{Sh } 36 = \textbf{Sh } 52 \text{ per composite unit.}$

03

Compute Composite Break-Even Units

$\text{Composite BEP} = \text{Total TFC} \div \text{WAC} = 120,000 \div 52 = \textbf{2,307.69 composite units.}$

04

Allocate by Product Mix

Product A: $2,307.69 \times 40\% = 923$ units. Product B: $2,307.69 \times 60\% = 1,385$ units. **This mix must be maintained for the BEP to hold.**

Worked Example: Product A & Product B — Base Data

The following two-product example is used throughout sections (i) to (ix). All subsequent calculations are based on these base assumptions.

Particulars	Product A	Product B
Selling Price per Unit (SP)	Sh 100	Sh 150
Variable Cost per Unit (VC)	Sh 60	Sh 90
Contribution per Unit (SP – VC)	Sh 40	Sh 60
Total Fixed Costs (TFC)	Sh 120,000	Sh 120,000
Fixed Cost Allocation Ratio	50%	50%
Fixed Costs Allocated	Sh 60,000	Sh 60,000
Tax Rate	30%	30%

(i) Computing Break-Even Units and Revenue

Formula: $Q_{BE} = TFC \div (SP - VC)$ | Break-Even Revenue = $Q_{BE} \times SP$

Calculation	Product A	Product B
Fixed Costs Allocated	Sh 60,000	Sh 60,000
SP per Unit	Sh 100	Sh 150
VC per Unit	Sh 60	Sh 90
Contribution per Unit	Sh 40	Sh 60
Break-Even Units	$60,000 \div 40 = \mathbf{1,500 \text{ units}}$	$60,000 \div 60 = \mathbf{1,000 \text{ units}}$
Break-Even Revenue	$1,500 \times 100 = \mathbf{Sh 150,000}$	$1,000 \times 150 = \mathbf{Sh 150,000}$

❏ **Conclusion:** Product A breaks even at 1,500 units and Sh 150,000 revenue. Product B breaks even at 1,000 units and Sh 150,000 revenue. Both products generate the same break-even revenue despite different unit volumes, due to Product B's higher selling price.

(ii) Computing Target Units for a Target Profit Before Tax

Management wants to earn a **target Profit Before Tax (PBT)** of Sh 20,000 from each product.

Formula: Target Units = (TFC + Target PBT) ÷ Contribution per Unit

Calculation	Product A	Product B
Fixed Costs Allocated	Sh 60,000	Sh 60,000
Target PBT	Sh 20,000	Sh 20,000
Required Contribution (TFC + PBT)	Sh 80,000	Sh 80,000
Contribution per Unit	Sh 40	Sh 60
Target Units (theoretical)	$80,000 \div 40 = 2,000$	$80,000 \div 60 = 1,333.33$
Practical Whole Units	2,000 units	1,334 units

Conclusion: The business must sell approximately 2,000 units of Product A and 1,334 units of Product B to achieve the target PBT of Sh 20,000 per product.

(iii) Target Revenue from After-Tax Profit & Margin of Safety

Management wants to achieve a **Profit After Tax (PAT)** of Sh 14,000 from each product, with a tax rate of 30%.

Step 1: Convert PAT to PBT

Target PBT = Target PAT ÷ (1 – Tax Rate)

= 14,000 ÷ (1 – 0.30)

= 14,000 ÷ 0.70

= **Sh 20,000**

Step 2: Target Units

$Q = (\text{TFC} + \text{Target PBT}) \div \text{Contribution per Unit}$

Product A: $80,000 \div 40 = \mathbf{2,000 \text{ units}}$

Product B: $80,000 \div 60 = \mathbf{1,334 \text{ units}}$ (rounded)

Margin of Safety Calculation

Item	Prod A	Prod B
Target Units	2,000	1,334
Breakeven Units	1,500	1,000
MOS in Units	500	334
MOS %	25%	25%

MOS % = MOS Units ÷ Target Units × 100. Both products show a 25% margin of safety, meaning sales can fall 25% before a loss is incurred.

(iv) Statement of Profit or Loss

Using the theoretical target quantities of 2,000 units for Product A and 1,333.33 units for Product B (theoretical figures used for exact reconciliation):

Item	Product A (Sh)	Product B (Sh)
Target Quantity	2,000 units	1,333.33 units
Sales Revenue	200,000	200,000
Less: Variable Costs	(120,000)	(120,000)
Contribution	80,000	80,000
Less: Fixed Costs	(60,000)	(60,000)
Profit Before Tax	20,000	20,000
Less: Tax @ 30%	(6,000)	(6,000)
Profit After Tax	14,000	14,000

Sales Revenue – Variable Costs =
Contribution

Contribution – Fixed Costs = **Profit Before Tax**

Profit Before Tax – Tax = **Profit After Tax**

(v) Computing the Selling Price to Break Even

Given a known quantity, variable cost, and fixed cost, what selling price is required to exactly break even?

Given — Product A

- Quantity = 2,000 units
- Variable Cost per Unit = Sh 60
- Total Fixed Costs = Sh 60,000

Formula & Calculation

$$\begin{aligned} SP &= [TFC + (VC \times Q)] \div Q \\ &= [60,000 + (60 \times 2,000)] \div 2,000 \\ &= 180,000 \div 2,000 \\ &= \text{Sh } 90 \text{ per unit} \end{aligned}$$

Verification Statement

Item	Sh
Total Revenue (90 × 2,000)	180,000
Total Variable Costs (60 × 2,000)	(120,000)
Fixed Costs	(60,000)
Profit/Loss	0

Therefore, Product A must be sold at **Sh 90 per unit** to break even at 2,000 units. Note this is below the actual selling price of Sh 100, confirming a profit buffer exists.

(vi) Computing the Maximum Variable Cost to Break Even

Given a known quantity, selling price, and fixed cost, what is the maximum variable cost per unit that still allows the business to break even?

Given — Product A

- Quantity = 2,000 units
- Selling Price = Sh 100 per unit
- Total Fixed Costs = Sh 60,000

Formula & Calculation

$$\begin{aligned} VC &= [(SP \times Q) - TFC] \div Q \\ &= [(100 \times 2,000) - 60,000] \div 2,000 \\ &= 140,000 \div 2,000 \\ &= \text{Sh 70 per unit} \end{aligned}$$

Verification Statement

Item	Sh
Total Revenue (100 × 2,000)	200,000
Total Variable Costs (70 × 2,000)	(140,000)
Fixed Costs	(60,000)
Profit/Loss	0

The maximum variable cost that Product A can incur and still break even at 2,000 units is **Sh 70 per unit**. The actual VC is Sh 60, providing a Sh 10 buffer per unit.

(vii) Determining Maximum Total Fixed Costs to Break Even

Given a known quantity, selling price, and variable cost, what is the maximum total fixed cost the business can sustain and still break even?

Given — Product A

- Quantity = 2,000 units
- Selling Price = Sh 100 per unit
- Variable Cost = Sh 60 per unit

Formula & Calculation

$$\begin{aligned}\text{TFC} &= (\text{SP} \times \text{Q}) - (\text{VC} \times \text{Q}) \\ &= (100 \times 2,000) - (60 \times 2,000) \\ &= 200,000 - 120,000 \\ &= \text{Sh } 80,000\end{aligned}$$

Verification Statement

Item	Sh
Total Revenue (100 × 2,000)	200,000
Total Variable Costs (60 × 2,000)	(120,000)
Maximum TFC at Break-Even	(80,000)
Profit/Loss	0

Product A can sustain total fixed costs of **Sh 80,000** and still break even at 2,000 units. The actual TFC is Sh 60,000, providing a Sh 20,000 fixed cost buffer.

(viii) Breakeven When Fixed Costs Cannot Be Allocated

When fixed costs cannot be meaningfully allocated to individual products, use the **weighted average contribution method**.
Assumptions: Total TFC = Sh 120,000 | Product A: 40% of mix, contribution Sh 40/unit | Product B: 60% of mix, contribution Sh 60/unit



Particular	Product A	Product B	Total
Product Mix	40%	60%	100%
Weighted Avg Contribution	$(40\% \times 40) = \text{Sh } 16$	$(60\% \times 60) = \text{Sh } 36$	Sh 52
Composite BE Units $(120,000 \div 52)$	—	—	2,307.69
Break-Even Units (allocated)	923 units	1,385 units	2,308 units
Selling Price	Sh 100	Sh 150	—
Break-Even Revenue	Sh 92,308	Sh 207,692	Sh 300,000

Sensitivity, Scenario & Monte Carlo Analysis

Three analytical techniques extend breakeven analysis to account for uncertainty and changing business conditions:

Sensitivity Analysis

Examines how the break-even point changes when **one key variable at a time** is changed. Main question: "What happens to BEP if one factor changes?" Low complexity. Best for identifying which individual assumptions have the greatest effect.

Scenario Analysis

Examines BEP under **different combinations of assumptions** representing possible situations (Best Case, Base Case, Worst Case). Several variables change simultaneously. Moderate complexity. Best for assessing alternative business conditions.

Monte Carlo Simulation

Uses **probability distributions and repeated simulations** (e.g., 10,000 runs) to estimate a range of possible break-even outcomes. Multiple uncertain variables change simultaneously. High complexity. Best for assessing risk where several variables are uncertain at once.

Comparing the Three Analytical Approaches

Aspect	Sensitivity Analysis	Scenario Analysis	Monte Carlo
Variables considered	One at a time (e.g., SP, VC, or TFC)	Several simultaneously (price, volume, costs)	Multiple uncertain variables with probability distributions
Treatment of uncertainty	Limited; focuses on impact, not probability	Defined assumptions; no explicit probabilities	Explicitly incorporates probabilities
Output	"A 10% rise in VC increases BEP units by 15%"	Best=5,000; Base=7,000; Worst=10,000 units	Probability distribution of BEP outcomes
Complexity	Low	Moderate	High
Best use	Identifying most impactful individual assumptions	Assessing alternative business conditions	Risk assessment with multiple simultaneous uncertainties

(ix) Product A — Sensitivity & Scenario Analysis

Base Case: SP = Sh 100 | VC = Sh 60 | TFC = Sh 60,000 | Contribution = Sh 40 | BEP = 1,500 units

Scenario	SP	VC	TFC	BEP Units	BEP Revenue
Base Case	100	60	60,000	1,500	150,000
(a) SP increases 10%	110	60	60,000	1,200	132,000
(b) VC increases 10%	100	66	60,000	1,765	176,471
(c) TFC increases 10%	100	60	66,000	1,650	165,000
(d) SP, VC & TFC +10%	110	66	66,000	1,500	165,000

(ix) Product B — Sensitivity & Scenario Analysis

Base Case: SP = Sh 150 | VC = Sh 90 | TFC = Sh 60,000 | Contribution = Sh 60 | BEP = 1,000 units

Scenario	SP	VC	TFC	BEP Units	BEP Revenue
Base Case	150	90	60,000	1,000	150,000
(a) SP increases 10%	165	90	60,000	800	132,000
(b) VC increases 10%	150	99	60,000	1,176	176,471
(c) TFC increases 10%	150	90	66,000	1,100	165,000
(d) SP, VC & TFC +10%	165	99	66,000	1,000	165,000

Summary of Sensitivity Results — Key Interpretations

Selling Price Increases by 10% → BEP Units Decrease

Higher selling prices increase contribution per unit, reducing the number of units required to cover fixed costs. This is the most favorable lever for management to pull.

Variable Cost Increases by 10% → BEP Units Increase

Higher variable costs reduce contribution per unit, increasing the sales volume required to break even. Cost control is therefore critical to maintaining a low BEP.

Fixed Costs Increase by 10% → BEP Units Increase by 10%

More fixed costs require additional contribution to reach break-even. The relationship is directly proportional a 10% rise in TFC causes exactly a 10% rise in BEP units.

SP, VC & TFC All Increase by 10% → BEP Units Unchanged

When all three variables increase proportionately, contribution per unit and fixed costs increase in tandem, leaving the break-even quantity unchanged — though BEP revenue rises.

Complete Formula Reference Sheet

All key formulas used in the two-product breakeven example, consolidated for quick reference:

Calculation	Formula
Profit Before Tax	$PBT = (SP \times Q) - (VC \times Q) - TFC$
Contribution per Unit	$Contribution = SP - VC$
Break-Even Units	$QBE = TFC \div (SP - VC)$
Break-Even Revenue	$BE\ Revenue = QBE \times SP$
Target PBT Units	$Q = (TFC + Target\ PBT) \div (SP - VC)$
Target PBT from PAT	$PBT = PAT \div (1 - Tax\ Rate)$
Selling Price at BE	$SP = [TFC + (VC \times Q)] \div Q$
Variable Cost at BE	$VC = [(SP \times Q) - TFC] \div Q$
Fixed Costs at BE	$TFC = (SP \times Q) - (VC \times Q)$
Weighted Average Contribution	$WAC = \Sigma(Product\ Mix\ \% \times Contribution\ per\ Unit)$
Composite Break-Even Units	$QBE = Total\ TFC \div Weighted\ Average\ Contribution$
Margin of Safety (units)	$MOS = Target\ Sales\ Units - Break-Even\ Units$
Margin of Safety (%)	$MOS\% = MOS\ Units \div Target\ Sales\ Units \times 100$

Challenges of Breakeven Analysis

Constant Price and Cost Assumption

Assumes constant selling price and variable cost per unit, which may not hold when prices, input costs, or discounts change with volume. In reality, bulk discounts, price negotiations, and market fluctuations are common.

All Output Is Sold Assumption

Assumes all output is sold, ignoring the possibility of unsold inventory, changing demand, or capacity constraints. Businesses routinely face stockpiling, seasonal demand, and supply chain disruptions.

Multi-Product Complexity

Difficult to apply where a business has multiple products, because different products have different selling prices, costs, and contribution margins. Allocation of fixed costs can be arbitrary and contentious.

Fixed Costs May Not Stay Fixed

Often assumes fixed costs remain constant, yet fixed costs may increase when production exceeds certain capacity levels (stepped fixed costs), making the model less reliable at high output volumes.

Short-Term, Simplified View

Provides a simplified, short-term view of profitability. It may not adequately capture uncertainty, competition, changes in technology, inflation, and other external factors that affect long-run business performance.

Contribution: The Heart of Breakeven Analysis

What Is Contribution?

Contribution per unit = Selling Price – Variable Cost per Unit. It represents the amount each unit sold contributes toward covering fixed costs. Once all fixed costs are covered, every additional unit of contribution becomes profit.

Why It Matters

Contribution is the central metric in all breakeven calculations. A higher contribution per unit means fewer units need to be sold to break even, and each additional unit sold beyond BEP generates more profit.

Contribution in Context

Low Contribution:

High variable costs relative to selling price. Business needs high volume to cover fixed costs. Vulnerable to cost increases.

High Contribution:

Low variable costs relative to selling price. Fewer units needed to break even. More resilient to fixed cost increases.

Contribution Margin Ratio: $\text{Contribution} \div \text{SP} \times 100$. Expresses contribution as a percentage of revenue useful for comparing products with different price points.

Margin of Safety — Measuring Business Risk

25%

MOS — Both Products

Both Product A and Product B show a 25% margin of safety when targeting Sh 14,000 PAT, meaning sales can fall 25% before a loss is incurred.

500

MOS Units — Product A

Target sales of 2,000 units minus break-even of 1,500 units = 500 units of safety buffer before losses begin.

334

MOS Units — Product B

Target sales of 1,334 units minus break-even of 1,000 units = 334 units of safety buffer before losses begin.

The Margin of Safety (MOS) measures how far actual or target sales can fall before the business begins to make a loss. It is expressed in units, revenue, or as a percentage of target sales. A higher MOS percentage indicates a more resilient business with greater protection against downturns in demand. Management should aim to maximize MOS by increasing contribution (through higher prices or lower variable costs) and controlling fixed costs.

The Breakeven Point and Business Risk

The breakeven point is fundamentally a measure of **business risk**. The higher the BEP relative to actual or expected sales, the greater the risk of making a loss.

Lower BEP = Lower Risk

The business reaches profitability sooner. Smaller sales volumes are needed to cover costs. Greater resilience to demand fluctuations and economic downturns.

Higher BEP = Higher Risk

The business must work harder selling more units or generating more revenue just to avoid a loss. Less margin for error in pricing, cost management, or demand forecasting.



Higher SP Reduces BEP

Increases contribution per unit, so fewer units are needed to cover fixed costs.



Higher VC Increases BEP

Reduces contribution per unit, requiring more units to cover the same fixed costs.



Higher TFC Increases BEP

More fixed costs require a greater total contribution before the business breaks even.

Breakeven Analysis in Pricing Decisions

Setting a Minimum Price

The breakeven selling price formula — $SP = [TFC + (VC \times Q)] \div Q$ — gives management the **minimum price** at which a product can be sold without incurring a loss at a given volume. Any price above this level generates profit; any price below it generates a loss.

In the Product A example, the breakeven SP at 2,000 units is **Sh 90**, while the actual SP is Sh 100 — providing a Sh 10 per unit pricing buffer.

Pricing Strategy Implications

- **Penetration pricing:** Setting price near BEP to gain market share, accepting thin margins initially
- **Premium pricing:** Setting price well above BEP to maximize contribution and reduce risk
- **Competitive pricing:** Using BEP as a floor and market rates as a ceiling
- **Tender pricing:** Calculating the minimum bid price that covers all costs at expected volume

Breakeven Analysis for Service Organizations

Breakeven analysis is equally applicable to service organizations, though the unit of measurement changes from physical products to service units.



Transport (Airlines/Buses)

Unit = passenger carried or seat-kilometre. BEP = minimum passengers to cover fixed costs (aircraft lease, crew salaries, terminal fees) given variable costs per passenger (fuel, catering, handling).



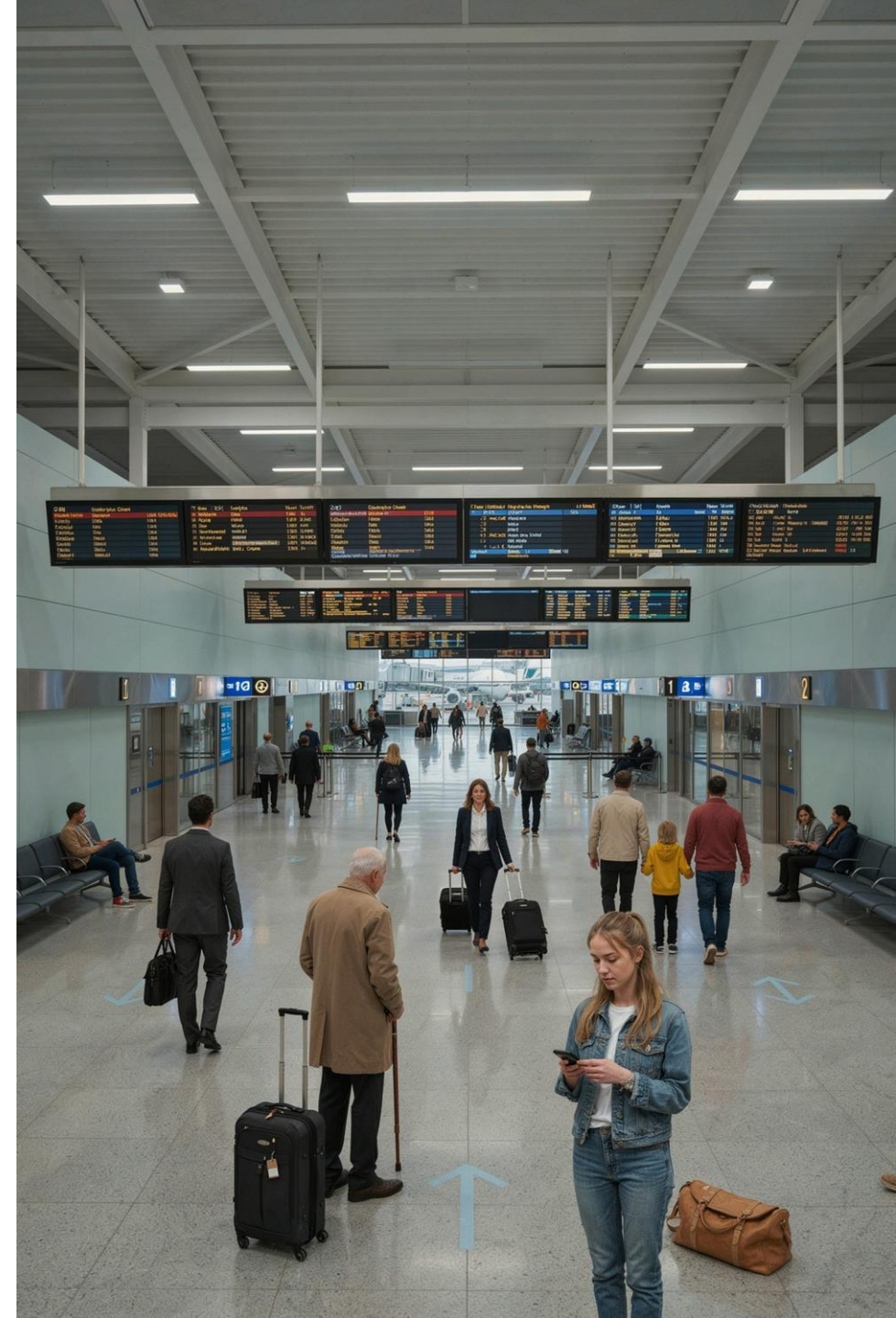
Healthcare

Unit = patient treated or bed-day. BEP = minimum patient volume to cover fixed costs (facility, equipment, permanent staff) given variable costs per patient (consumables, drugs, variable staffing).



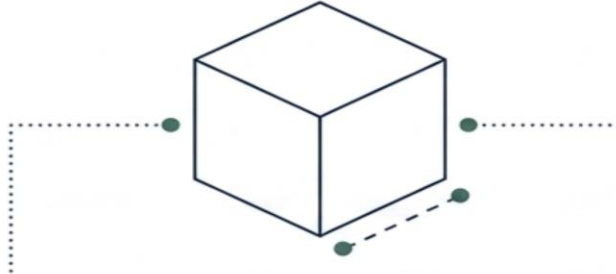
Hospitality

Unit = room-night or cover (restaurant). BEP = minimum occupancy rate or covers to cover fixed costs (property, permanent staff, utilities) given variable costs per guest (laundry, food, amenities).

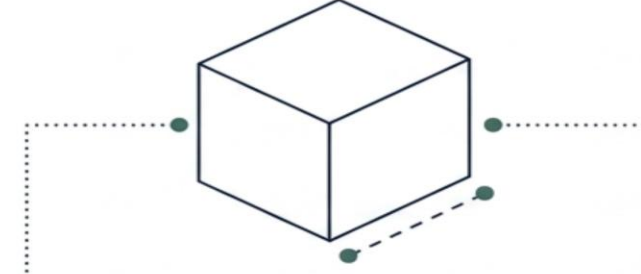


Sensitivity Analysis — The Impact of a 10% Change

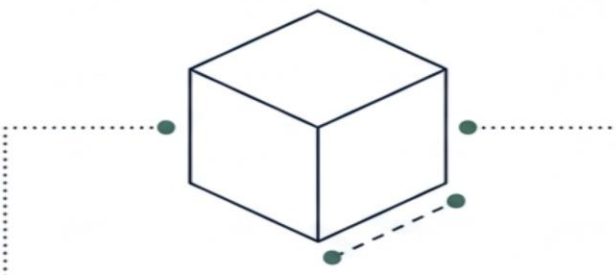
A structured sensitivity analysis reveals which variable has the greatest impact on the break-even point.
The results from the Product A and B analysis are summarized below:



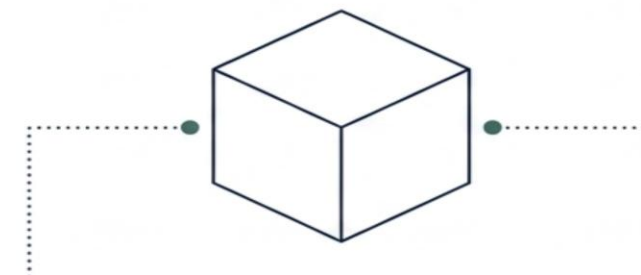
SP +10% causes BEP units to decrease — favorable



VC +10% causes BEP units to increase — unfavorable



TFC +10% causes BEP units to increase by exactly 10% — proportional



SP, VC and TFC all +10% causes BEP units unchanged — neutral

The key insight is that **selling price increases** are the **most powerful lever** for reducing BEP, while variable cost increases are the most damaging. When all three variables increase proportionately, the BEP quantity is unchanged though BEP revenue rises because the higher selling price means each unit generates more revenue.

Scenario Analysis — Best, Base, and Worst Case

Scenario analysis extends sensitivity analysis by changing multiple variables simultaneously to represent coherent business situations. Unlike sensitivity analysis (one variable at a time), scenario analysis tests realistic combinations of conditions.

Best Case Scenario

SP increases 10%, VC decreases 5%, TFC remains constant. Higher contribution per unit and unchanged fixed costs produce the **lowest possible BEP**. Represents favorable market conditions: strong demand, efficient production, stable overhead.

Base Case Scenario

All variables at their current/expected levels. This is the **planning baseline** used for budgeting and target-setting. Product A BEP = 1,500 units; Product B BEP = 1,000 units at base assumptions.

Worst Case Scenario

SP decreases 10%, VC increases 10%, TFC increases 10%. Contribution per unit falls sharply while fixed costs rise, producing the **highest possible BEP**. Represents adverse conditions: price competition, input cost inflation, capacity expansion.

Monte Carlo Simulation in Breakeven Analysis

Monte Carlo simulation is the most sophisticated approach to breakeven analysis under uncertainty. Rather than testing one scenario at a time, it runs **thousands of simulations** simultaneously, each drawing random values for SP, VC, and TFC from specified probability distributions.

How It Works

- Assign probability distributions to each uncertain variable (e.g., SP is normally distributed with mean Sh 100 and standard deviation Sh 10)
- Run 10,000 simulations, each randomly sampling from these distributions
- Calculate the BEP for each simulation
- Aggregate results into a probability distribution of BEP outcomes

Output and Interpretation

The output is a **probability distribution** showing the range and likelihood of different BEP outcomes. For example: "There is a 90% probability that the BEP will fall between 1,200 and 1,800 units." This enables management to make risk-informed decisions rather than relying on a single-point estimate.

Monte Carlo is particularly valuable when multiple variables are uncertain simultaneously and their interactions are complex.

Break-even Analysis and Profit Planning

From BEP to Target Profit

Once the break-even point is established, management can extend the analysis to determine the sales volume required to achieve any desired profit level. The formula $Q = (TFC + \text{Target PBT}) \div (SP - VC)$ is the core tool for profit planning.

This allows management to set **sales targets** that are grounded in financial reality rather than arbitrary aspirations.

Practical Profit Planning Steps

1. Determine the target profit (after tax if applicable)
2. Gross up PAT to PBT using: $PBT = PAT \div (1 - \text{Tax Rate})$
3. Add TFC to target PBT to get required total contribution
4. Divide by contribution per unit to get target sales volume
5. Multiply by SP to get target revenue
6. Calculate MOS = Target Units – BEP Units

Limitations of the Breakeven Assumptions — A Critical View

Constant Prices Rarely Hold

In practice, selling prices change with volume (bulk discounts, promotional pricing), market conditions (competition, inflation), and customer negotiations. The assumption of a single constant SP oversimplifies real-world pricing dynamics.

Inventory Exists in Reality

The assumption that all output is sold ignores the reality of opening and closing inventory, seasonal demand patterns, and production lead times. Unsold inventory means fixed costs are not fully recovered in the period.

Fixed Costs Are Stepped, Not Flat

Fixed costs often increase in steps when production exceeds capacity thresholds, for example, hiring additional supervisors, leasing extra warehouse space, or purchasing new equipment. The model assumes a flat fixed cost line, which is unrealistic at high output levels.

External Factors Are Ignored

The model does not capture competition, technological change, inflation, regulatory changes, or macroeconomic conditions all of which can materially affect costs, prices, and demand in ways that invalidate the breakeven calculation.

Overcoming Breakeven Limitations in Practice



Use Sensitivity Analysis

Test how BEP changes as individual assumptions are relaxed. Identify the most critical variables and focus management attention on controlling them.



Build Multiple Scenarios

Develop Best Case, Base Case, and Worst Case scenarios that reflect realistic combinations of market conditions, rather than relying on a single-point estimate.



Apply Monte Carlo Simulation

Where multiple variables are uncertain, use simulation to generate a probability distribution of BEP outcomes and make risk-informed decisions.



Use Weighted Average Contribution

For multi-product businesses, apply the weighted average contribution method to handle fixed cost allocation challenges and compute a composite BEP that reflects the actual product mix.

Key Formulas at a Glance — Quick Reference

Break-Even Units

$$Q_{BE} = TFC \div (SP - VC)$$

Break-Even Revenue

$$BE\ Rev = Q_{BE} \times SP$$

Target Units (PBT)

$$Q = (TFC + PBT) \div (SP - VC)$$

PAT to PBT

$$PBT = PAT \div (1 - \text{Tax Rate})$$

Margin of Safety

$$MOS = \text{Target Units} - BEP\ \text{Units}$$

MOS Percentage

$$MOS\% = MOS \div \text{Target Units} \times 100$$

SP at Break-Even

$$SP = [TFC + (VC \times Q)] \div Q$$

VC at Break-Even

$$VC = [(SP \times Q) - TFC] \div Q$$

TFC at Break-Even

$$TFC = (SP \times Q) - (VC \times Q)$$

Weighted Avg Contribution

$$WAC = \Sigma(\text{Mix}\% \times \text{Contribution/Unit})$$

Composite BEP

$$Q_{BE} = \text{Total TFC} \div WAC$$

Contribution per Unit

$$\text{Contribution} = SP - VC$$

Topic 7 Summary — Breakeven Analysis

01

The Core Concept

BEP is where Total Revenue = Total Costs. The root formula $\Pi = (SP \times Q) - (VC \times Q) - TFC$ can be rearranged to solve for any unknown variable.

02

Key Metrics

Break-even units ($Q_{BE} = TFC \div \text{Contribution}$), break-even revenue, target sales volume, and margin of safety are the four primary outputs of C-V-P analysis.

03

Multi-Product Situations

Allocate fixed costs where possible; use weighted average contribution and composite BEP where allocation is not feasible. Always maintain the assumed product mix.

04

Uncertainty Management

Use sensitivity analysis (one variable), scenario analysis (multiple variables), and Monte Carlo simulation (probability distributions) to extend BEP analysis beyond its deterministic assumptions.

05

Limitations

BEP assumes constant prices, no inventory, constant fixed costs, and a single product mix. These simplifications must be recognized and addressed through the analytical techniques above.